



$$\begin{matrix} & \begin{matrix} X_1 & X_2 \end{matrix} \\ \begin{matrix} X_1 \\ X_2 \end{matrix} & \begin{bmatrix} .8 & .4 \\ .2 & .6 \end{bmatrix} \end{matrix} = \begin{bmatrix} P(X_t = X_1 | X_{t-1} = X_1) & P(X_t = X_1 | X_{t-1} = X_2) \\ P(X_t = X_2 | X_{t-1} = X_1) & P(X_t = X_2 | X_{t-1} = X_2) \end{bmatrix}$$

$$P(X_t = X_1) = P$$

$$P(X_t = X_2) = P(X_{t-1} = X_1) P(X_t = X_2 | X_{t-1} = X_1) + P(X_{t-1} = X_2) P(X_t = X_2 | X_{t-1} = X_2)$$

$$\begin{bmatrix} P(X_t = X_1) \\ P(X_t = X_2) \end{bmatrix} = \begin{bmatrix} P(X_t = X_1 | X_{t-1} = X_1) & P(X_t = X_1 | X_{t-1} = X_2) \\ P(X_t = X_2 | X_{t-1} = X_1) & P(X_t = X_2 | X_{t-1} = X_2) \end{bmatrix} \begin{bmatrix} P(X_{t-1} = X_1) \\ P(X_{t-1} = X_2) \end{bmatrix}$$

$$P(X_t) = A P(X_{t-1}) = A^t P(X_{t_0})$$

$$P(X_{t+1}) = A P(X_{t_0})$$

$$P(X_{t+2}) = A P(X_{t+1}) = A A P(X_{t_0})$$

$$P(X = X_1) = 0.8 P(X = X_1) + 0.4 P(X = X_2)$$

$$P(X = X_2) = 0.2 P(X = X_1) + 0.6 P(X = X_2)$$

$$P(X_T = X_2) = 2 P(X_T = X_2) \rightarrow P(X_2) = \frac{1}{2} P(X_1)$$

$$P(X_1) + P(X_2) = 1$$

$$P(X_1) + \frac{1}{2} P(X_1) = 1 \rightarrow \begin{matrix} P(X_1) = 2/3 \\ P(X_2) = 1/3 \end{matrix}$$

$$P(X_1) = 0.8 \frac{2}{3} + 0.4 \frac{1}{3} = \frac{8 \cdot 2 + 4}{30} = \frac{20}{30} = \frac{2}{3}$$

$$P(X_2) = 0.2 \frac{2}{3} + 0.6 \frac{1}{3} = \frac{2 \cdot 2 + 6}{30} = \frac{10}{30} = \frac{1}{3}$$

Los sistemas de recomendación funcionan así, las mas probables son las primeras

- $s \mapsto (s_1, \dots, s_n)$ estado $|S|$

- $A = \{a_1, \dots, a_m\}$

- acciones: $S \rightarrow P(A)$
- acciones(s) $\subseteq A$

- $\left[\begin{array}{l} \text{sucesor: } S \times A \rightarrow S \\ \text{sucesor}(s, a) \in S \end{array} \right] \rightarrow S \times A \times S \rightarrow \mathbb{R} \quad \begin{array}{l} \text{sucesor}(s, a, s') \\ P(s_{T+1} = s' | s_T = s, A_t = a) \end{array}$

- $\left[\begin{array}{l} \text{costo_local } S \times A \rightarrow \mathbb{R}^+ \\ \text{costo_local}(s, a) \in \mathbb{R}^+ \end{array} \right] \rightarrow \text{Recompensa: } S \times A \times S \rightarrow \mathbb{R} \quad \text{recompensa}(s, a, s')$

- $\left[\begin{array}{l} \text{Terminal } S \rightarrow \{F, V\} \\ \text{terminal}(s) \end{array} \right]$